SOC HOME LAB

Wazuh Configuration and Endpoint Onboarding

A concise technical overview of the lab architecture, core platform, and Windows agent integration.

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Project purpose

This project demonstrates how an open-source security monitoring stack can be assembled in a virtualized lab. Wazuh provides centralized event analysis and endpoint monitoring, while Suricata and pfSense contribute network security visibility. Kali Linux is used as a controlled test system for generating lab activity.

Environment note: This guide is written for a Windows endpoint target. The source notes reference Windows 11 Pro; the lab documentation should reflect the version actually deployed. If using the planned Windows 10 VM, record it as Windows 10.

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01 | Lab Overview

The Wazuh SOC Lab brings together endpoint and network telemetry to create a small, practical Security Operations Center environment. The goal is to centralize security events, identify suspicious activity, and provide a foundation for later detection engineering and incident investigation.

Core components

Component	Role in the lab
Wazuh Server	Hosts the Wazuh Manager, Indexer, and Dashboard. Receives endpoint and integrated security telemetry, analyzes events, and supports search and visualization.
Windows endpoint	Runs the Wazuh Agent to collect host security events and other configured endpoint data. Use the actual installed Windows version in the final record.
Kali Linux	Controlled attacker/test machine used to generate activity for monitoring and detection exercises.
pfSense	Provides firewall and network event data that can add context to investigations and support anomaly monitoring.
Suricata IDS/IPS	Inspects network traffic and generates security alerts that can be forwarded to Wazuh.

Architecture flow: endpoint, firewall, and IDS/IPS telemetry is collected or forwarded to the Wazuh platform. The Manager decodes and evaluates events, the Indexer stores searchable data, and the Dashboard presents alerts and investigation views. Insert the project's architecture image here in a future revision if available.

02 | SIEM and Wazuh Fundamentals

A Security Information and Event Management (SIEM) platform aggregates logs and security events from endpoints, servers, applications, users, and network infrastructure. By making this data searchable and correlating activity across sources, a SIEM helps analysts monitor environments, identify suspicious patterns, and investigate potential incidents.

Wazuh is an open-source security platform used in this lab for threat detection, file integrity monitoring, security configuration assessment, compliance-related visibility, and response workflows. Its accessibility and customization make it suitable for a cost-conscious learning environment.

Wazuh platform architecture

- **Agents** collect configured endpoint telemetry from Windows, Linux, and macOS systems and send it to the Wazuh server.
- **Wazuh Manager** receives and analyzes events using decoders and rules, producing alerts when configured conditions match.
- **Wazuh Indexer** indexes and stores event data to support search and analytics.
- **Wazuh Dashboard** provides a browser-based interface for monitoring, alert review, querying, and investigation.

In this lab, the Wazuh server is deployed as a virtual machine and accessed through its web dashboard. The remaining components contribute additional endpoint and network visibility as they are integrated.

03 | Deployment Environment and Access

The current configuration notes identify the following environment:

Setting	Value
Host platform	VMware Workstation / VMware virtualization
Virtual network mode	NAT (Network Address Translation)
Target agent operating system	Windows (document the exact version used)
Wazuh access	HTTPS to the Wazuh VM's IP address

Deployment approach

Wazuh offers a virtual machine appliance in OVA format, allowing the platform to be imported into supported virtualization software instead of installing every server component individually. After the VM starts, identify its network address and open the dashboard over HTTPS. Credentials should be taken from the current appliance documentation or the lab's secure setup notes.

Official virtual machine deployment documentation: Wazuh Virtual Machine Deployment

Security note: Do not publish default credentials, active passwords, private keys, or sensitive IP information in a public portfolio. Change default credentials where applicable and keep the lab isolated from untrusted networks.

04 | Core Service Health and Dashboard Readiness

If the dashboard displays a “Server not ready” message, verify that the main Wazuh services are active before changing configuration. The Manager processes and analyzes events, the Indexer stores searchable data, and the Dashboard serves the user interface. A failure or startup delay in any of these services can prevent the interface from becoming ready.

Service checks

Run these commands in the Wazuh VM terminal with appropriate administrative privileges:

```
sudo systemctl status wazuh-manager
sudo systemctl status wazuh-indexer
sudo systemctl status wazuh-dashboard
```

If a service is confirmed inactive, start the relevant service and re-check its status:

```
sudo systemctl start wazuh-manager
sudo systemctl start wazuh-indexer
sudo systemctl start wazuh-dashboard
```

Once all required services report as running, refresh the dashboard. If the problem remains, review the relevant service logs and startup errors rather than repeatedly restarting services without checking their state.

The original setup notes use `sudo su` to obtain a root shell. Prefer running individual commands with `sudo` when practical, as this limits the time spent in a privileged shell.

05 | Windows Agent Onboarding and Validation

The Wazuh Agent is a lightweight endpoint component that forwards configured security data to the Manager. Depending on the operating system and configuration, agent capabilities can include log collection, File Integrity Monitoring (FIM), Security Configuration Assessment (SCA), and malware-related detection.

Agent onboarding summary

- Obtain the Windows agent MSI from the official Wazuh documentation and install it on the target Windows VM.
- In the Wazuh Dashboard, use the endpoint deployment workflow to define the Wazuh Manager address and a unique agent name (for example, `PC1`).
- Run the dashboard-generated registration and setup commands in an elevated PowerShell session on the Windows endpoint.
- Return to the dashboard and confirm that the new agent appears and reports an **Active** status.

Official Windows agent package and installation documentation: Wazuh Agent for Windows

Validation checklist

Validation item	Expected result
Agent registration	The endpoint is listed in the Wazuh Dashboard with the intended agent name.
Agent connectivity	The agent status is Active and it can communicate with the Manager.
Telemetry	Relevant Windows events appear in Wazuh after the configured data sources generate activity.
Evidence	Capture a sanitized screenshot of the active agent and note the test date and endpoint OS version.

On a typical 64-bit Windows installation, agent files are commonly located under `C:\Program Files (x86)\ossec-agent`. Confirm the actual installation path on the endpoint. The agent installation includes service and configuration files; use the installed agent's supported management options to inspect logs or control the service.

06 | Operational Notes and Next Steps

A working dashboard and an Active agent establish the initial monitoring baseline. The next stage is to confirm that useful endpoint events are arriving, then integrate pfSense and Suricata so investigations can use both host and network evidence.

- Record the actual VM names, operating system versions, network mode, and component versions.
- Keep evidence reproducible: include the test objective, expected result, observed result, and timestamp.
- Use controlled lab activity only, and retain sanitized screenshots or logs that demonstrate a detection or investigation.
- Document any issue encountered, the diagnostic evidence, the change made, and the validation performed afterward.

Reference links

- Wazuh virtual machine deployment
- Wazuh Windows agent installation

Portfolio documentation standard: This guide describes the intended setup and validation criteria. Mark an item as completed only after it has been tested in the actual lab, and replace generic descriptions with your own sanitized evidence as the project progresses.